AI Jobs Analytics Dashboard – Data Visualization

Project Overview

The **AI Jobs Analytics Dashboard** is a data visualization project that analyzes trends in the global Artificial Intelligence job market. The objective of this project is to understand how AI adoption is influencing job opportunities, salary levels, work-life balance, career growth, and industry investment across different countries.

Artificial Intelligence has become one of the fastest-growing technology sectors. Many industries are adopting AI solutions to improve efficiency, automate processes, and make better decisions using data. As a result, new job roles such as AI Engineer, Machine Learning Engineer, Data Scientist, and Computer Vision Engineer have emerged.

This project focuses on analysing these roles and identifying patterns related to job demand, salary distribution, career growth, and job security using an interactive dashboard.

The dashboard allows users to explore the data through filters such as **job role, year, and work mode**, making the analysis dynamic and interactive.`

Tools Used

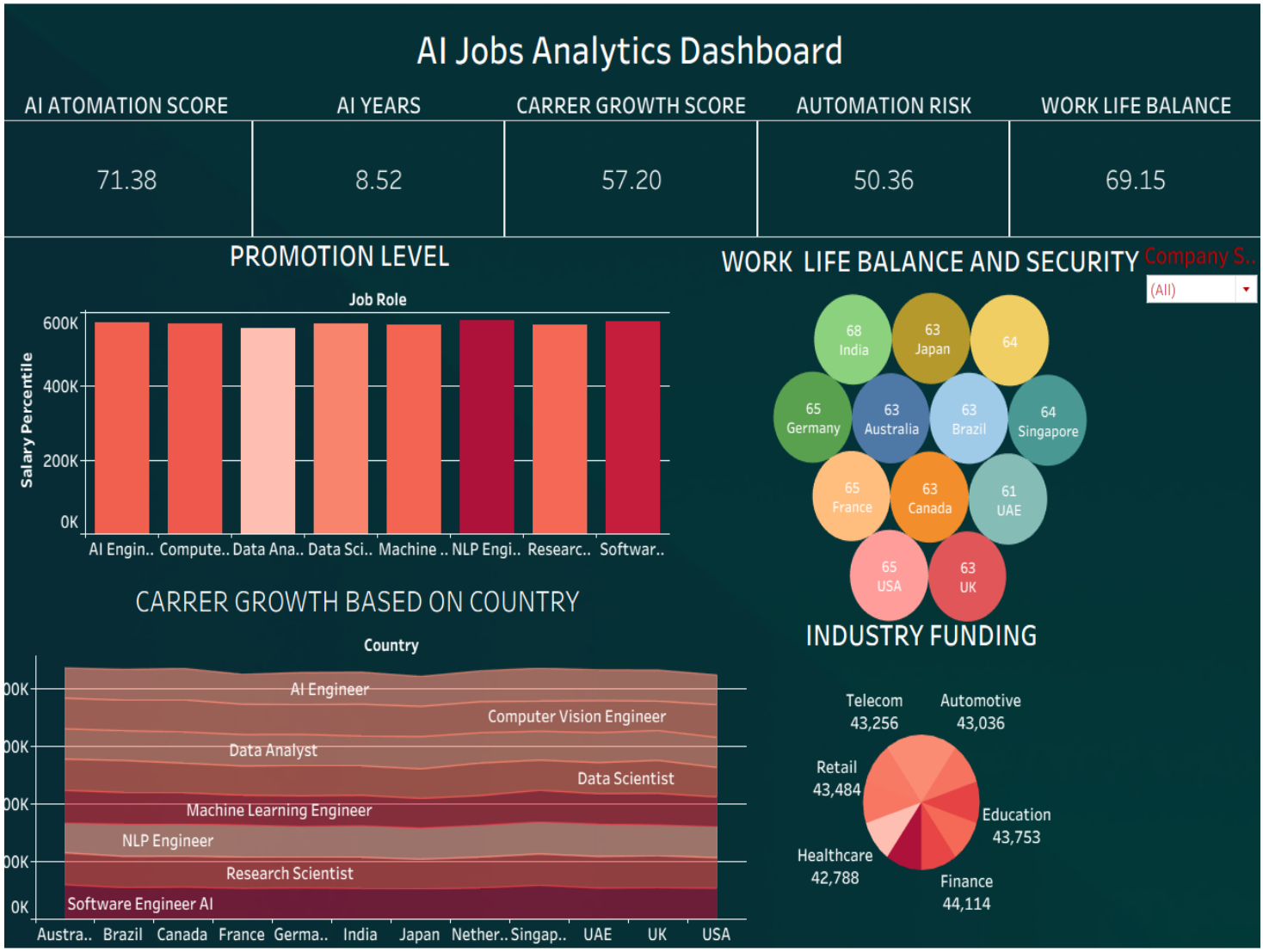
Excel

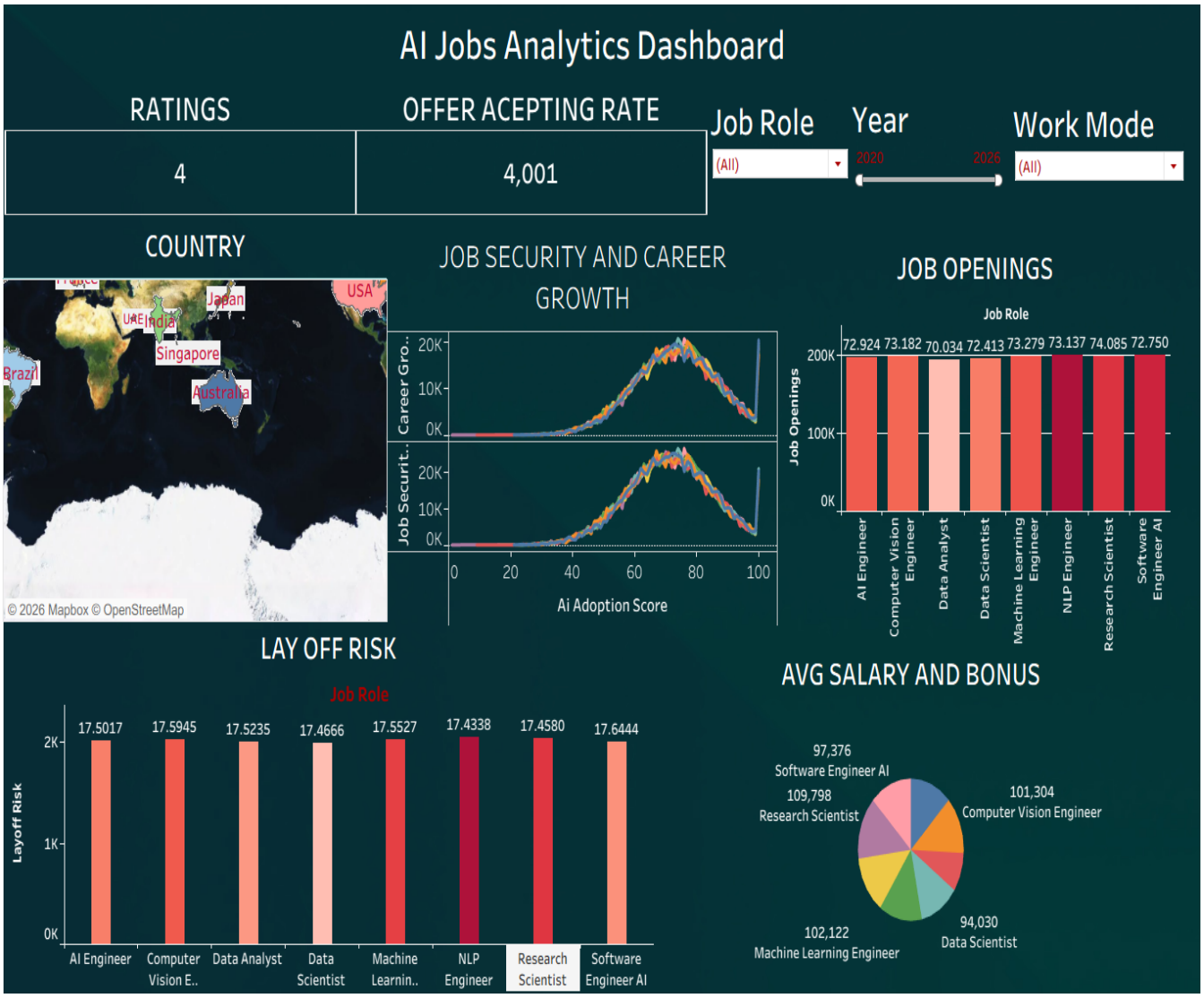
Excel was used for **data preparation and cleaning**. Tasks included formatting the dataset, organizing columns, and ensuring that the data was ready for visualization.

Tableau

Tableau was used to create the **interactive dashboard and visualizations**. Tableau helped transform the dataset into visual insights using different chart types such as bar charts, pie charts, maps, and line graphs.

Dashboard Components and Analysis





1. Key Performance Indicators (KPIs)

At the top of the dashboard, several KPI indicators provide a quick overview of the dataset.

AI Automation Score

The AI Automation Score measures the level of AI adoption across industries. A score of **71.38** suggests that many organizations are actively implementing AI technologies.

AI Years

This metric represents the average number of years of experience required in AI-related roles. The dashboard shows **8.52 years**, indicating that many AI positions require significant expertise and specialized skills.

Career Growth Score

The career growth score reflects how quickly professionals can advance in AI-related careers. The value **57.20** indicates strong opportunities for career development.

Automation Risk

Automation risk indicates the possibility of job roles being replaced by automated systems. A score of **50.36** shows moderate risk, meaning that while AI can automate certain tasks, it also creates new job opportunities.

Work-Life Balance

The work-life balance score of **69.15** indicates that many AI professionals experience relatively good balance between work responsibilities and personal life.

2. Promotion Level by Job Role

This bar chart compares salary percentile levels across different AI-related job roles.

The roles included in the analysis are:

- AI Engineer
- Computer Vision Engineer
- Data Analyst
- Data Scientist
- Machine Learning Engineer
- NLP Engineer
- Research Scientist

- Software Engineer (AI)

The chart shows that most AI roles have high salary percentiles, indicating strong demand and competitive compensation.

Machine Learning Engineers and Research Scientists appear to have slightly higher promotion potential due to their advanced technical expertise.

3. Work-Life Balance and Security

This bubble chart visualizes work-life balance scores across different countries.

Countries included in the analysis:

- India
- USA
- UK
- Germany
- Japan
- Australia
- Brazil
- France
- Canada
- Singapore
- UAE

Each bubble represents a country, and the size and color reflect its work-life balance score.

The visualization shows that some countries such as **India and Germany** have relatively high scores, while others maintain moderate levels due to competitive work environments.

4. Career Growth Based on Country

This stacked area chart shows how different AI job roles grow across various countries.

Countries with advanced technology ecosystems such as:

- USA
- Germany
- Singapore
- Canada

show stronger career growth opportunities.

The visualization also shows that AI adoption is expanding globally, creating new opportunities even in emerging markets.

5. Industry Funding

This pie chart displays the distribution of AI investment across different industries.

Industries included in the analysis are:

- Finance
- Education
- Telecom
- Retail
- Automotive
- Healthcare

The chart shows that **Finance and Education sectors receive significant investment in AI technologies**. This is because organizations in these sectors rely heavily on predictive analytics, automation, and intelligent systems.

Healthcare is also rapidly adopting AI for diagnostics and medical research.

6. Country Distribution Map

The map visualization highlights countries where AI jobs are concentrated.

Countries such as:

- USA

- India
- Japan
- Singapore
- Australia

appear prominently on the map.

This indicates that AI development and job opportunities are distributed across multiple regions globally.

7. Job Security and Career Growth

This line chart shows the relationship between AI adoption and career growth.

The graph indicates that as AI adoption increases, career growth opportunities also increase. This suggests that professionals with AI-related skills will remain in high demand in the future.

8. Job Openings by Role

This bar chart compares job openings across different AI job roles.

The analysis shows that:

- AI Engineers
- Machine Learning Engineers
- Data Scientists

have the highest number of job openings.

This indicates that organizations are actively recruiting professionals with advanced AI and data analysis skills.

9. Layoff Risk

This chart analyzes the layoff risk associated with different AI job roles.

The values across roles are relatively similar, suggesting that AI-related careers generally provide stable employment compared to other technology fields.

10. Average Salary and Bonus

This pie chart compares average salaries among different AI job roles.

The results show that:

- Research Scientists
- Machine Learning Engineers

receive higher salaries due to their advanced research skills and specialized knowledge.

Data Scientists and Computer Vision Engineers also receive competitive compensation.

Key Insights

The dashboard analysis reveals several important insights:

- AI job demand is increasing across multiple industries.
- Machine Learning Engineers and AI Engineers have the highest demand.
- Countries with strong technology ecosystems offer better career growth opportunities.
- AI adoption is directly related to job market expansion.
- Industries such as finance and healthcare are investing heavily in AI technologies.

Conclusion

The **AI Jobs Analytics Dashboard** provides a comprehensive analysis of the global AI job market. The visualizations reveal how AI adoption is shaping career opportunities, salary levels, and industry investments.

As artificial intelligence continues to evolve, the demand for AI professionals will grow significantly. Professionals who develop skills in machine learning, data science, and AI engineering will have strong career prospects in the future.

This project demonstrates the power of data visualization in transforming complex datasets into meaningful insights that help understand real-world trends.

main dashboard used for analysis in this project.